

A take-away game uses one pile of counters and a set of allowed moves, such as {1, 3, 4}. Two players take turns. On each turn a player removes a number of counters that is in the set, and no more than the pile has. Whoever takes the last counter wins.

A pile is a **losing pile** (L) if the player about to move will lose when the other player plays carefully. Every other pile is a **winning pile** (W). A pile of 0 is a losing pile, because the other player has just taken the last counter.

Problem 1: Allowed moves {1, 2}. Describe all the losing piles, and explain why your description is correct for piles of every size.

Problem 2: Allowed moves {1, 3, 4}. Label each pile from 0 to 29 with W or L. Describe all the losing piles, and explain why your description is correct for piles of every size.

0	1	2	3	4	5	6	7	8	9

10	11	12	13	14	15	16	17	18	19

20	21	22	23	24	25	26	27	28	29

Problem 3: Allowed moves $\{1, 3, 4\}$. A player always takes as many counters as the rules allow. This player moves first, and the opponent plays carefully. From which piles does this player win?

Problem 4: Allowed moves $\{1, 3, 5\}$. Describe all the losing piles. Find every set of allowed moves that contains 1 and has exactly the same losing piles as $\{1, 3, 5\}$.

Problem 5: Allowed moves $\{1, 2, m\}$, where m is a whole number larger than 2. For which m are the losing piles exactly the multiples of 3, as they are for $\{1, 2\}$? For every other m , describe the losing piles.

Problem 6: Allowed moves $\{1, 6, 9\}$. Label each pile from 0 to 39 with W or L. From some pile on, the labels repeat in a cycle forever. Find the pile where the repeating starts and the length of the cycle.

0	1	2	3	4	5	6	7	8	9

10	11	12	13	14	15	16	17	18	19

20	21	22	23	24	25	26	27	28	29

30	31	32	33	34	35	36	37	38	39

Problem 7: Explain why, for every finite set of allowed moves, the labels W and L eventually repeat in a cycle forever.

In a game with two piles, a player chooses one pile on each turn and removes an allowed number of counters from that pile only. Whoever takes the last counter wins.

Give each single pile a **value**. A pile of 0 has value 0. The value of any other pile is the smallest whole number that is not the value of a pile you can reach from it in one move.

Problem 8: Allowed moves {1, 3, 4}. Find the value of each pile from 0 to 20.

0	1	2	3	4	5	6	7	8	9

10	11	12	13	14	15	16	17	18	19

20

Problem 9: Allowed moves $\{1, 3, 4\}$, with two piles. For each pair of piles, decide whether the player about to move can win. When that player can win, find a winning first move.

piles of **6** and **4**

piles of **5** and **1**

piles of **12** and **5**

piles of **11** and **3**

Problem 10: Explain why, with two piles, the player about to move loses against careful play when the two values are equal, and can win when the two values are different.